

PROTEIN CHEMISTRY & PEPTIDES

Definitions, peptides, classifications, functions and levels of protein structure

1. Definition of proteins

Proteins are nitrogen-containing, high molecular weight organic macromolecules composed of amino acids. In biochemical terms, proteins are polymers of L-alpha-amino acids joined together by covalent peptide bonds.

Proteins form the fundamental basis of cellular structure and function. Their properties depend on the number, type, sequence and spatial arrangement of amino acids in the polypeptide chain.

A polypeptide is usually called a protein when it is large enough to fold into a definite functional structure. Many proteins may contain one or more polypeptide chains.

2. Peptides

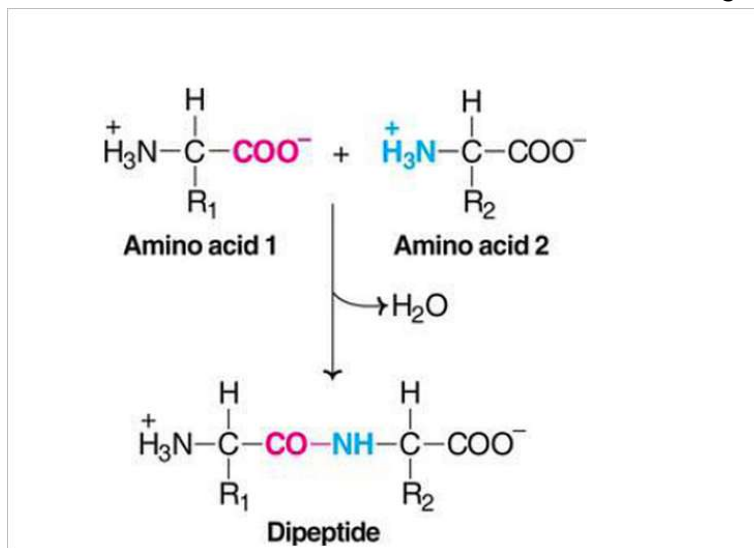
Definition of peptides

Peptides are short chains of amino acids joined by peptide bonds. The source text generally applies the term peptide when the number of constituent amino acids is less than 10.

A dipeptide has 2 amino acids and 1 peptide bond. A tripeptide has 3 amino acids and 2 peptide bonds. A decapeptide has 10 amino acids and 9 peptide bonds. Peptides containing more than 10 amino acids are usually referred to as polypeptides.

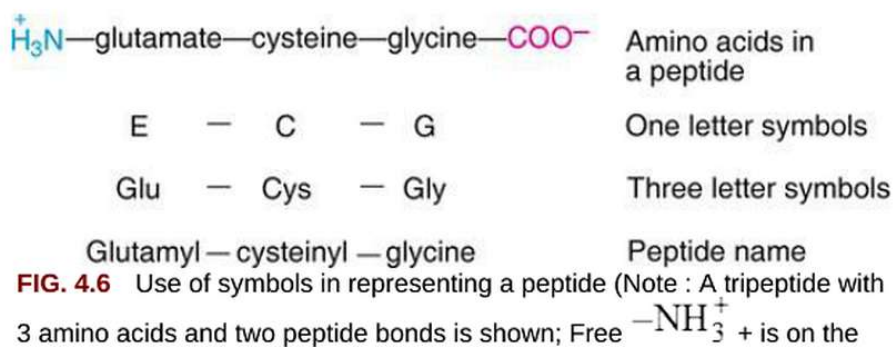
Peptide bond

A peptide bond is formed when the carboxyl group (-COOH) of one amino acid combines with the amino group (-NH₂) of another amino acid, with removal of water. The resulting linkage is -CO-NH-.



Source Fig. 4.5: Peptide bond formation from the provided Biochemistry PDF.

- Peptide bonds are strong covalent bonds and form the backbone of protein structure.
- The peptide bond is rigid and planar because it has partial double-bond character.
- The peptide chain is written from the N-terminal end on the left to the C-terminal end on the right.
- The amino acid sequence is read from N-terminal to C-terminal direction.



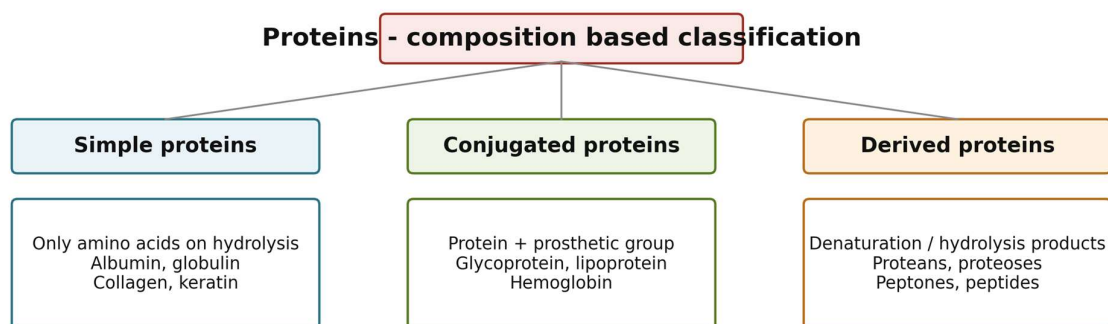
Source Fig. 4.6: Symbols and naming of a tripeptide from the provided source.

3. Biologically important peptides

Peptide	Nature / composition	Biological importance
Glutathione	Tripeptide: gamma-glutamyl-cysteinyl-glycine	Antioxidant defense; maintains RBC membrane integrity; protects hemoglobin; helps detoxication and amino acid transport.
Thyrotropin releasing hormone (TRH)	Tripeptide	Hypothalamic hormone that stimulates release of thyrotropic hormone from pituitary gland.
Oxytocin	Nonapeptide	Posterior pituitary hormone; causes contraction of uterus and helps milk ejection.
Vasopressin / ADH	Nonapeptide	Promotes water retention by kidneys; helps increase blood pressure.
Angiotensin I and II	Angiotensin I: decapeptide; Angiotensin II: octapeptide	Angiotensin II has strong hypertensive effect and stimulates aldosterone release.
Methionine and leucine enkephalins	Pentapeptides	Opioid peptides in brain; inhibit pain sensation and affect emotions.
Bradykinin and kallidin	Bradykinin: nonapeptide; kallidin: decapeptide	Powerful vasodilator peptides formed from plasma proteins.
Peptide antibiotics	Peptide molecules	Examples: gramicidin, bacitracin, tyrocidin and actinomycin.
Aspartame	Dipeptide derivative	Artificial sweetener; about 200 times sweeter than sucrose.
Gastrointestinal peptide hormones	Peptide hormones	Examples: gastrin and secretin; regulate gastrointestinal secretion and movement.

4. Classification of proteins based on composition

Based on chemical composition and products of hydrolysis, proteins are classified into simple proteins, conjugated proteins and derived proteins.



Use this chart for quick recall: composition asks what the protein yields on hydrolysis.

Revision chart: Composition-based classification of proteins.

A. Simple proteins

Simple proteins are composed only of amino acid residues. On hydrolysis, they yield only amino acids.

Class	Key features	Examples
Albumins	Soluble in water and dilute salt solutions; coagulated by heat	Serum albumin, ovalbumin, lactalbumin
Globulins	Soluble in neutral and dilute salt solutions	Serum globulins, vitelline
Glutelins	Soluble in dilute acids and alkalies; mainly plant proteins	Glutelin of wheat, oryzenin of rice
Prolamines	Soluble in 70% alcohol	Gliadin of wheat, zein of maize
Histones	Strongly basic proteins associated with DNA	Thymus histones
Protamines	Strongly basic, smaller than histones; associated with nucleic acids	Sperm proteins
Fibrous proteins	Long, fiber-like, insoluble, resistant to digestion	Collagen, elastin, keratin

B. Conjugated proteins

Conjugated proteins contain a protein part plus a non-protein part called a prosthetic group.

Type	Prosthetic group	Examples
Nucleoproteins	Nucleic acid	Nucleohistones, nucleoprotamines
Glycoproteins / mucoproteins	Carbohydrate	Mucin, ovomucoid, immunoglobulins
Lipoproteins	Lipid	Serum lipoproteins, chylomicrons
Phosphoproteins	Phosphoric acid	Casein, vitelline
Chromoproteins	Coloured group / pigment	Hemoglobin, cytochromes
Metalloproteins	Metal ions such as Fe, Cu, Zn or Mg	Ceruloplasmin, carbonic anhydrase

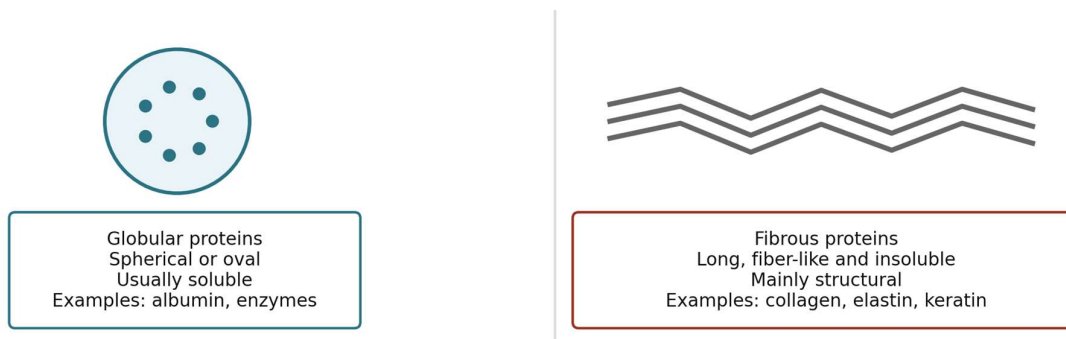
C. Derived proteins

Derived proteins are produced by denaturation, coagulation or hydrolysis of simple or conjugated proteins.

Group	Meaning	Examples
Primary derived proteins	Early products formed by denaturation or first stage of hydrolysis	Coagulated proteins, proteans, metaproteins
Secondary derived proteins	Further hydrolytic products produced by progressive breakdown of peptide bonds	Proteoses, peptones, polypeptides, peptides

5. Classification of proteins based on shape

Classification based on shape



Revision chart: Globular and fibrous proteins.

Shape class	Features	Examples
Globular proteins	Spherical or oval; usually soluble; mostly dynamic proteins	Albumins, globulins, enzymes, hemoglobin
Fibrous proteins	Long, fiber-like, insoluble; mostly structural and resistant to digestion	Collagen, elastin, keratin

6. Classification of proteins based on function

Functional classification of proteins

Structural	Catalytic	Transport
collagen, keratin	enzymes	hemoglobin, albumin
Hormonal	Contractile	Storage
insulin, GH	actin, myosin	ovalbumin, ferritin
Defense	Receptor	Genetic
immunoglobulins	membrane receptors	histones, protamines

Static functions = structure/support. Dynamic functions = enzymes, transport, hormones, movement and defense.

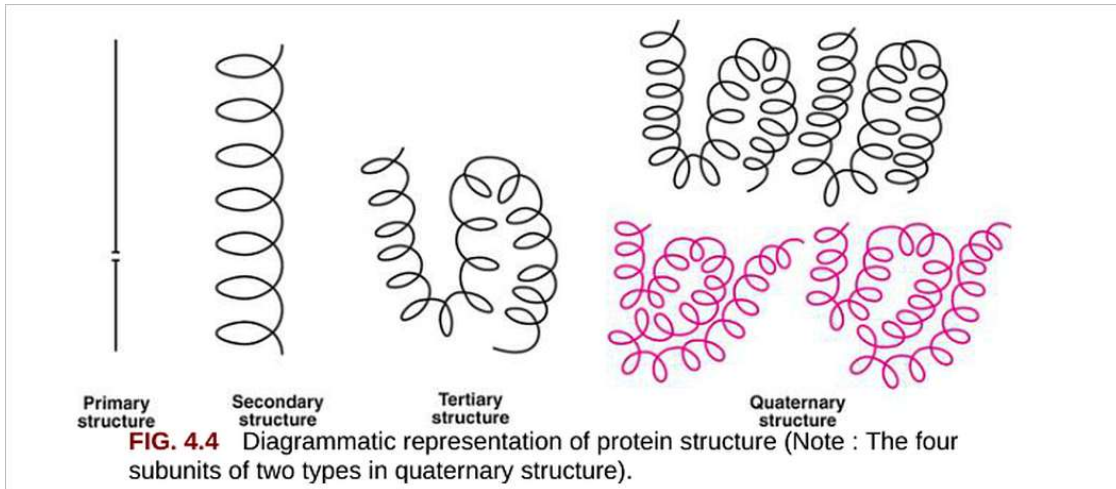
Revision chart: Functional classification of proteins with examples.

Function class	Main role	Examples
Structural	Provide strength, support and shape to tissues	Collagen, keratin, elastin
Catalytic / enzymes	Accelerate biochemical reactions	Hexokinase, pepsin, DNA polymerase
Transport	Carry substances in blood or across membranes	Hemoglobin, albumin, transferrin
Hormonal / regulatory	Regulate metabolism and physiological activities	Insulin, growth hormone
Contractile	Produce movement and muscle contraction	Actin, myosin
Storage	Store amino acids or other molecules	Ovalbumin, ferritin, casein
Defense	Protect body against infection or bleeding	Immunoglobulins, fibrinogen

Function class	Main role	Examples
Receptors	Receive signals and mediate cell response	Hormone receptors, membrane receptors
Genetic / nuclear proteins	Associate with DNA and gene regulation	Histones, protamines

7. Levels of protein structure

Proteins are not just straight chains of amino acids. Their biological activity depends on the correct organization of the polypeptide chain into four levels: primary, secondary, tertiary and quaternary structure.



Source Fig. 4.4: Primary, secondary, tertiary and quaternary levels of protein structure from the provided PDF.

A. Primary structure

Primary structure means the linear sequence of amino acids in a polypeptide chain. This sequence forms the backbone of the protein and is held together by peptide bonds.

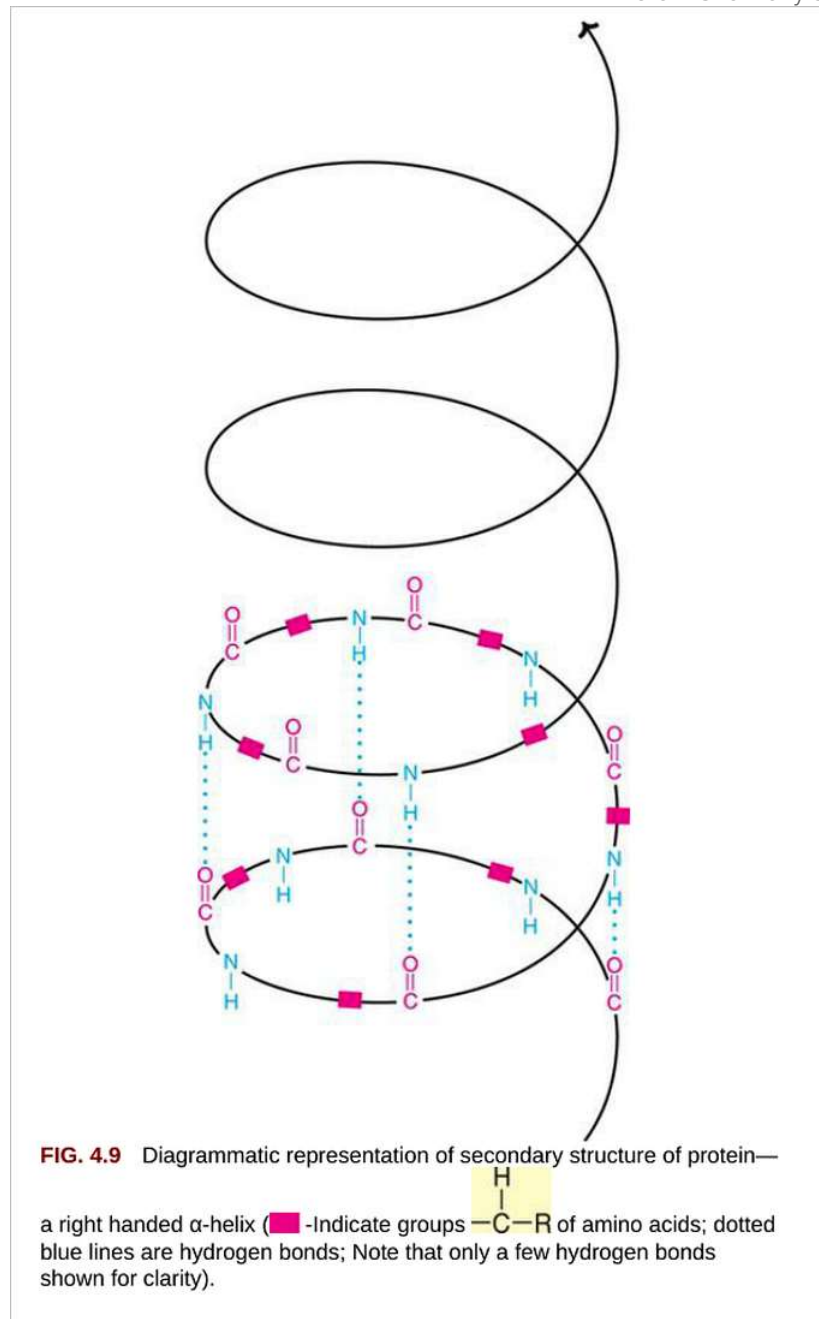
- Each protein has a unique amino acid sequence determined by genes in DNA.
- Primary structure is largely responsible for protein function because it determines how the chain will fold.
- Abnormal amino acid sequence can cause disease. Example: a single amino acid substitution in hemoglobin causes sickle-cell hemoglobin.
- Primary structure is written from the N-terminal amino acid to the C-terminal amino acid.

B. Secondary structure

Secondary structure refers to local folding or twisting of the polypeptide chain. It mainly includes alpha-helix and beta-pleated sheet. These structures are stabilized mainly by hydrogen bonds between peptide bond C=O and N-H groups.

Alpha-helix

The alpha-helix is a common spiral arrangement of the polypeptide chain. It is a tightly coiled structure with amino acid side chains projecting outward. The source describes it as the most common spiral structure of protein.

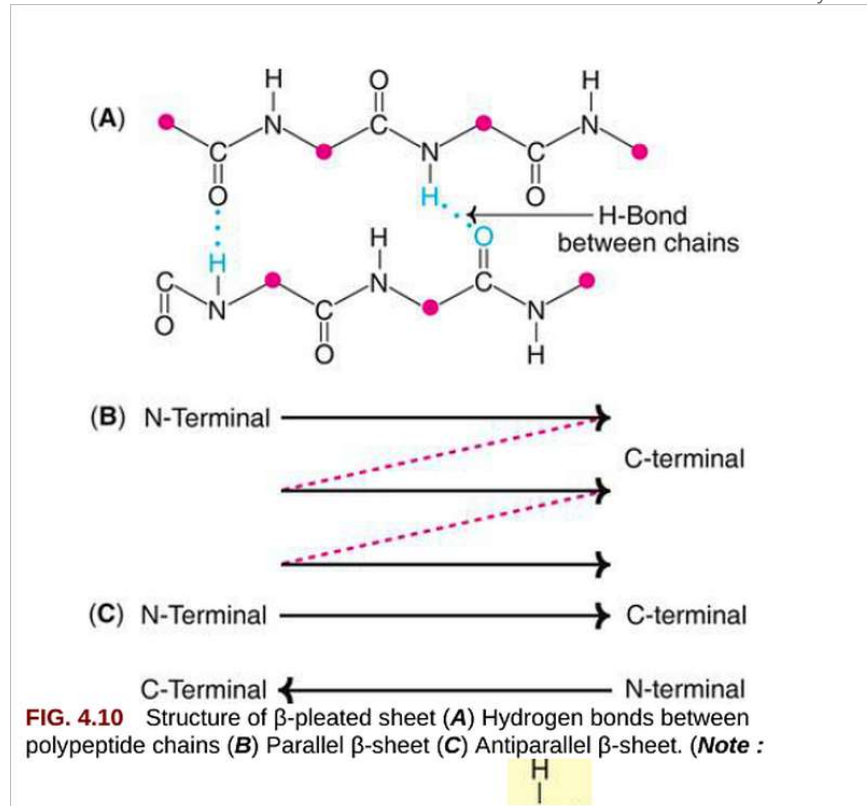


Source Fig. 4.9: Right-handed alpha-helix showing hydrogen bonds.

- Hydrogen bonds stabilize the helix.
- Each turn of alpha-helix contains about 3.6 amino acids.
- Proline and many acidic or basic amino acids may disturb alpha-helix formation.

Beta-pleated sheet

Beta-pleated sheets are formed by two or more segments of fully extended polypeptide chains. Hydrogen bonds are formed between neighbouring segments. The chains may be parallel or antiparallel.



Source Fig. 4.10: Beta-pleated sheet with hydrogen bonding and parallel/antiparallel arrangement.

C. Tertiary structure

Tertiary structure is the three-dimensional folding of a single polypeptide chain into a compact and biologically active form. It converts the secondary structural elements into a functional protein molecule.

- It is stabilized by hydrophobic interactions, hydrogen bonds, ionic bonds, van der Waals forces and disulfide bonds.
- Tertiary structure forms important functional regions such as enzyme active sites and binding sites.
- Examples: myoglobin and many enzymes are functional mainly at tertiary level.
- Denaturation usually disrupts tertiary structure and causes loss of biological activity.

D. Quaternary structure

Quaternary structure is the spatial arrangement of two or more polypeptide chains, called subunits, in a protein molecule. Each subunit may have its own primary, secondary and tertiary structure.

- Only proteins having more than one polypeptide chain show quaternary structure.
- Subunits may be identical or different.
- Examples: hemoglobin has four subunits, two alpha and two beta chains; immunoglobulins also have multiple polypeptide chains.
- Quaternary structure is important for cooperative functions, regulation and stability of many proteins.

8. Functions of proteins in the body

- **Structural support:** Collagen, elastin and keratin provide strength to bone, cartilage, tendons, skin, hair and nails.
- **Catalysis:** Most enzymes are proteins and speed up metabolic reactions such as digestion, glycolysis and DNA replication.
- **Transport:** Hemoglobin transports oxygen; albumin transports fatty acids, bilirubin and drugs; transferrin transports iron.

- **Storage:** Ferritin stores iron; casein and ovalbumin store amino acids.
- **Hormonal regulation:** Insulin, growth hormone, oxytocin and vasopressin regulate body functions.
- **Movement:** Actin and myosin are responsible for muscle contraction and cellular movement.
- **Defense:** Immunoglobulins protect against infection, and clotting proteins prevent blood loss.
- **Receptors and signaling:** Membrane proteins act as receptors and allow cells to respond to hormones, neurotransmitters and other signals.
- **Buffering and fluid balance:** Plasma proteins help maintain blood pH and colloid osmotic pressure.
- **Energy source:** During starvation, amino acids from proteins can be used for energy or converted into glucose.

9. Quick exam summary

Term	High-yield point
Protein	Polymer of L-alpha amino acids joined by peptide bonds.
Peptide	Short chain of amino acids; usually fewer than 10 amino acids in the source text.
Peptide bond	Covalent -CO-NH- linkage between carboxyl group of one amino acid and amino group of another.
Composition classification	Simple, conjugated and derived proteins.
Shape classification	Globular and fibrous proteins.
Functional classification	Structural, enzymatic, transport, hormonal, contractile, storage, defense, receptor and genetic proteins.
Protein structure levels	Primary sequence, secondary folding, tertiary 3D structure, quaternary subunit arrangement.
Key examples	Glutathione, oxytocin, vasopressin, angiotensin, enkephalins, collagen, hemoglobin, albumin, insulin.